

Fast, Stable, and Efficient Joint Modeling Approaches in Rare Diseases Settings

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Abstract

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1 Introduction

Especially when sample sizes are small, it is important to increase information-content and hence efficiency by collecting multiple outcomes per study participant, typically measured longitudinally over time. Such studies may or may not include one or several time-to-event outcomes as well. This is often referred to as joint modelling and/or multivariate longitudinal outcomes (Verbeke *et al.* 2014; Cadarso *et al.* 2017).

Unsurprisingly, such articulated data against the background of relatively small sample sizes tend to lead to complex parameter estimation and subsequent inference problems, often based on intractable likelihoods. Considerable work has been done to improve fitting and convergence properties. Pseudo-likelihood offers a convenient way out (Geys, Molenberghs, and Lipsitz 1998; Geys, Molenberghs and Ryan 1997). The basic principle is that a complex likelihood is replaced by an alternative function that still leads to consistent and asymptotically normal estimators, with a small amount of statistical efficiency loss, but important gains in computational properties (reduced computation time; improved convergence properties). Renard *et al.* (2004) applied pseudo-likelihood to alleviate computational complexity in a multilevel probit model setting. The survival setting was examined by Tibaldi *et al.* (2004). Pseudo-likelihood has also been adapted to work when data are incomplete (Parzen *et al.* 2007, Molenberghs *et al.* 2011).

Whereas pseudo-likelihood is still based on a single objective function, Fieuws and Verbeke (2006) proposed so-called pairwise likelihood, where a potentially large collection of longitudinal sequences is analyzed by first analyzing every possible pair of sequences separately, after which combination rules are applied to yield a single set of inferences. Fieuws *et al.* (2008) used multivariate longitudinal data for prediction purposed in patients with renal grafts. Delporte *et al.* (2025b) started from this technique to accelerate convergence. Delporte *et al.* (2025a) presented an application in high-dimensional longitudinal data. De Witte *et al.* (2025) used pairwise modeling ideas to efficiently jointly analyze many longitudinal sequences and a time-to-event outcome. The specific case of combined continuous and binary responses was studied by Delporte *et al.* (2022, 2025c).

Molenberghs, Verbeke, and Iddi (2011) proposed a so-called split-sample technique to improve computational convenience in very large datasets. While this is seemingly the opposite of the situation arising in rare diseases, the technique has potential as well when sample sizes are limited. Indeed, while the analysis of longitudinal and hierarchical data is stabler when data are balanced (e.g., all longitudinal sequences are equally long, or all clusters have the same size) and even admit closed-form solutions (Flórez Poveda *et al.* 2019ab, 2022), this is no longer true in the realistic settings with unbalanced data. Hermans *et al.* (2018ab) proposed two-stage approaches where a sample is partitioned according to cluster sizes. Each part is then analyzed separately, after which a weighted combination is applied to produce a single estimator. Molenberghs *et al.* (2014) provided some theoretical underpinning. Alternatively, clusters of unequal sizes, such as trials in a meta-analysis, can be analyzed more efficiently by applying multiple imputation with the sole purpose of producing clusters of equal size (Van der Elst *et al.* 2016).

Alonso *et al.* (2015) proposed a simulation-based method when a large group of raters evaluated a large group of compounds.

Evidently, different computational devices can be combined to enhance efficiency further. For example, Ivanova, Molenberghs, and Verbeke (2016, 2017ab) applied both pairwise-likelihood and split-sampling techniques, in particular in the context of multivariate longitudinal ordinal data, and reached a much better computational efficiency than when only one of these two techniques was applied. An overview is provided of a number of devices to stabilize and/or speed up computation towards estimation and inference.

2 Methodology Overview

This is now a sketch, leading up to the simulation study. At a later stage, it will be turned in a genuine section, describing the finally chosen methodology.

The proposal is to focus on a combination of the methodology of Hermans (#514), Flórez Poveda (#546 and #622), and Van der Elst (#471):

- Hermans studied the use of weighting methods in the context of clustered data with

unequal size. The limitation is that Hermans focused attention on compound-symmetry models. This has the advantage of allowing for explicit expressions for several weighting schemes. The disadvantage is that some of these weighting schemes do not easily generalize to more general mixed models.

- Flórez Poveda considered a closed-form expression for a general class of mixed models (but not all fit into them). The limitation is that the data need to be balanced (e.g., no missing data or no clusters of unequal size for other reasons).
- Van der Elst (#471) used multiple imputation to restore equality of cluster sizes.

The idea here is to combine the advantages of both. In other words, we will study rather general mixed models for continuous longitudinal or otherwise hierarchical data (e.g., clusters consisting of centers in a multi-center trial), where clusters are of unequal size. The generality is dictated by fitting into Flórez Poveda framework. It is similar to what is examined in #546. Indeed, in this paper, we studied models for the evaluation of surrogate endpoints; these are different from compound-symmetry models, but still of a very particular nature. Here, we want to allow for somewhat more general models.

For a given set of longitudinal or clustered data, with a corresponding linear mixed model formulated, we aim to do the following:

- Apply classical maximum likelihood
- Apply the closed-form plus weighting as in #568.
- Apply multiple imputation first, as in #471 and then apply the closed form of #546.

An extension will consist of multivariate longitudinal data, rather than, for example, a single sequence.

3 Simulation Study

We consider the following elements:

- Number of clusters or longitudinal sequences $N = 10, 000; 100; 50; 20; 10$ (the large set is to also examine the performance in relatively large samples, but also to see what the “asymptotic” bias is when comparing the alternative methods to full maximum likelihood).
- Number of repeated measures n_i is uniform between in $[1, 12]$
- A linear mixed model with fixed and random intercepts and slopes in time (can gradually be made more complex). The parameters can be varied across simulation settings, but as a proof-of-concept let us start with one set.

- Analysis using the aforementioned three methods

4 Concluding Remarks

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Acknowledgments

ERDERA, RealiseD

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